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Masterarbeit

ScholarCopilot with Fine-Grained Citation Retrieval

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Selbstständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Masterarbeit selbständig und nur mit den angegebenen Hilfsmitteln angefertigt habe und dass alle Stellen, die dem Wortlaut oder dem Sinne nach anderen Werken entnommen sind, durch Angaben von Quellen als Entlehnung kenntlich gemacht worden sind. Diese Masterarbeit wurde in gleicher oder ähnlicher Form in keinem anderen Studiengang als Prüfungsleistung vorgelegt.

Niklas Munkes (Matrikelnummer 4269436), August 23, 2025

Abstract

TODO

Contents

1 Introduction

TODO

2 Related Work

TODO

2.1 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG, [LPP+21]) models are designed to generate text that is grounded in real-world factual knowlege. They have been widely adopted for knowledge-intensive NLP tasks [SQK+25], such as long-form question answering [XWL+24], scientific question answering [LOS+23], generation with sentence-level citations [ZBL+24, XWL+24] and academic text generation [AHS+24, WMN+25]. RAG models are hybrid architectures that usually consist of a generator, a pre-trained sequence-to-sequence (seq2seq) model (e.g. a Transformer-based [VSP+17] decoder, see section 2.2), and a retriever, which may be implemented as a dense vector index (e.g. a HNSW [MY16] index, see section 3.1.1) [LPP+21]. Thus they combine pre-trained paramentric memory with non-paramentric (retrieval-based) memory [LPP+21]. Using an external

Previous work on text generation with citations has recently transitioned from treating retrieval and generation as seperate processing stages (e.g. [SHC+24]) to alternating between the two (e.g. [XWL+24], [AHS+24]). *Attribute First, then Generate* [SHC+24] initially retrieves a set of relevant sub-sentence spans which are then grouped into coherent clusters which in turn are used to inform an iterative sentence-by-sentence generation step [SHC+24]. *ReClaim* [XWL+24], more specifically the *ReClaim with Interleaving Generation* method, is used to generate an output sequence $O = \{r_1, c_1, ..., r_n, c_n\}$ of alternating references and claims where claim c_i is the portion of the models response that was generated based on reference r_i. Claim and reference generation is performed by seperate LLMs specifically trained for their corresponding task [XWL+24]. In contrast, *OpenScholar* [AHS+24] first retrieves and reranks a set of papers P relevant to the given query. These papers inform a subsequent generation step that produces a generated response r_i [AHS+24]. This response is refined through iterative *self-feedback* where during each iteration of the feedback loop $F = f_1, ..., f_T$, the generator model generates sentences f_j that describe potential improvements to the response and may also prompt the retrieval pipeline to retrieve additional papers, should r_i require more information to generate an improved response r_{i+1} [AHS+24]. Depending on the size of P and the choice of T, OpenScholar still performs retrieval and generation largely seperately.

ScholarCopilot [WMN+25] is a RAG system developed for generating academic-style text with accurate citations. It is intended as a tool that assists the user with academic paper writing by providing iterative text generation with automated citation retrieval [WMN+25]. ScholarCopilot also enables user interaction during generation [WMN+25] by allowing the user to rewrite the generated text or forcefully trigger citation retrieval between the consecutive generation steps but this feature is beyond the scope of this thesis and is thus not utilized for the experiments. ...

2.2 Decoder Architectures

TODO

3 Methods

TODO

3.1 ScholarCopilot

TODO

3.1.1 HNSW

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4 Experiments

TODO

5 Conclusion

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Bibliography

[AHS+24] Akari Asai, Jacqueline He, Rulin Shao, Weijia Shi, Amanpreet Singh, Joseph Chee Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D'arcy, David Wadden, Matt Latzke, Minyang Tian, Pan Ji, Shengyan Liu, Hao Tong, Bohao Wu, Yanyu Xiong, Luke Zettlemoyer, Graham Neubig, Dan Weld, Doug Downey, Wen tau Yih, Pang Wei Koh, and Hannaneh Hajishirzi. Openscholar: Synthesizing scientific literature with retrieval-augmented lms, 2024.

[LOS+23] Jakub Lála, Odhran O'Donoghue, Aleksandar Shtedritski, Sam Cox, Samuel G. Rodriques, and Andrew D. White. Paperqa: Retrieval-augmented generative agent for scientific research, 2023.

[LPP+21] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks, 2021.

[MY16] Yu. A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs, 2016.

[SHC+24] Aviv Slobodkin, Eran Hirsch, Arie Cattan, Tal Schuster, and Ido Dagan. Attribute first, then generate: Locally-attributable grounded text generation, 2024.

[SQK+25] Rulin Shao, Rui Qiao, Varsha Kishore, Niklas Muennighoff, Xi Victoria Lin, Daniela Rus, Bryan Kian Hsiang Low, Sewon Min, Wen tau Yih, Pang Wei Koh, and Luke Zettlemoyer. Reasonir: Training retrievers for reasoning tasks, 2025.

[VSP+17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017.

[WMN+25] Yubo Wang, Xueguang Ma, Ping Nie, Huaye Zeng, Zhiheng Lyu, Yuxuan Zhang, Benjamin Schneider, Yi Lu, Xiang Yue, and Wenhu Chen. ScholarCopilot: Training large language models for academic writing with accurate citations, 2025.

[XWL+24] Sirui Xia, Xintao Wang, Jiaqing Liang, Yifei Zhang, Weikang Zhou, Jiaji Deng, Fei Yu, and Yanghua Xiao. Ground every sentence: Improving retrieval-augmented llms with interleaved reference-claim generation, 2024.

[ZBL+24] Jiajie Zhang, Yushi Bai, Xin Lv, Wanjun Gu, Danqing Liu, Minhao Zou, Shulin Cao, Lei Hou, Yuxiao Dong, Ling Feng, and Juanzi Li. Longcite: Enabling llms to generate fine-grained citations in long-context qa, 2024.